Arrival · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Four checks. Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Fraction

What percentage is 1/4?

Help: Multiply by 100.

2. Genotypes

What is the genotype ratio for Aa × Aa?

Help: Count AA, Aa and aa.

3. Phenotypes

With complete dominance, what is the phenotype ratio?

Help: AA and Aa show the dominant phenotype.

4. Independence

Does an aa offspring force the next offspring to be AA?

Help: Each fertilisation is a new event.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W05-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Point, speak, sign, draw, write or direct an adult. An adult may read or scribe your exact words. Choose one route; do not complete all three.

1. Calculate

Complete the expected aa counts for A, B and C.

Help: Total × 0.25.

2. Convert

Convert observed 7/20 and 18/80 to percentages.

Help: Count ÷ total × 100.

3. Explain

Explain why 7/20 does not automatically disprove the one-in-four model.

Help: Expected proportions need not match small samples exactly.

My response: _____

Work as a pair. One explains a step. One asks for the clue. Swap roles. Keep your own first answer and your own next step.

Choose one route for the full 10 minutes.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit ticket · Independent

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

1. Calculate

Cross Aa × Aa. Find the expected aa count in 80 offspring.

Help: Find one quarter of 80.

2. Interpret

Why might a sample of 20 contain 7 aa rather than exactly 5?

Help: Small samples vary by chance.

My response: _____

Made from the web lesson SCI-LAUNCH-A2-W05-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Expected versus observed

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

All samples model Aa × Aa. Calculate the expected aa count, then compare it with the simulated result. Do not change the theoretical probability because of one sample.

Sample	Total	Expected aa	Observed aa
A	20	___	7
B	20	___	3
C	80	___	18

My evidence and reasoning

One limit or useful check

My response: _____

Keep your first answer. Hear the feedback. Choose one real change, or explain why to keep it. Ask for a read-back if it helps.

Made from the web lesson SCI-LAUNCH-A2-W05-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Punnett square · working sheet

Convert fractions to percentages and ratios.

Calculate expected numbers and compare observations.

Explain why each fertilisation is independent.

Record the parent alleles. Fill all four boxes. Then find the chance of aa.

Aa × Aa · side / top	**A**	**a**
A	_____	_____
a	_____	_____

Compare with Aa × aa.

Aa × aa · side / top	**A**	**a**
a	_____	_____
a	_____	_____

Aa × Aa chance of aa: ____ . Aa × aa chance of aa: ____ .

Made from the web lesson SCI-LAUNCH-A2-W05-L3
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale